Zero Phase Systems

Example Ideal LPF

$$H(\omega) = \begin{cases} 1 ; & |\omega| < \omega_0 \\ 0 ; & \omega_0 \le |\omega| \le \pi \end{cases}$$

Impulse response

$$h(n) = \frac{\sin \omega_0 n}{\pi n}, \quad n \in Z$$

$$\phi(\omega) = \angle H(\omega) = 0 \quad \text{for all } \omega$$

Group delay

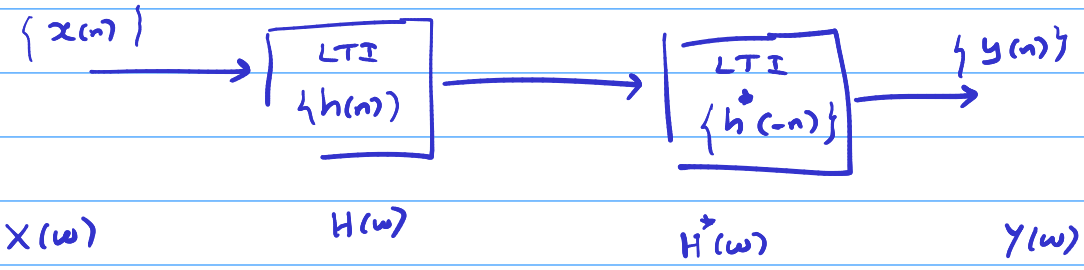
$$\tau(\omega) = 0 \quad \text{for all } \omega.$$

IIR

Non causal

Not realizable in practice.

Another way to get zero phase



$$Y(\omega) = \underbrace{|H(\omega)|^2}_{(\text{real})} X(\omega)$$

Zero Phase System

If $\{h(n)\}$ is causal then $\{h^*(-n)\}$ will be anti-causal

except for trivial case $h(n) = \alpha \delta(n)$

Zero phase system $\Rightarrow$ Not causal

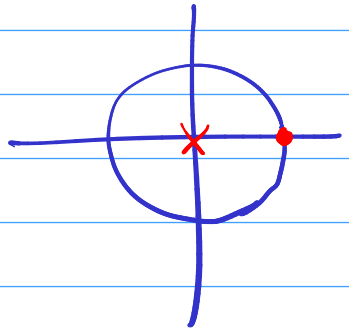
Linear Phase Systems.

Example:

$$H(z) = 1 - z^{-1}$$

$$\downarrow z = e^{j\omega}$$

$$H(\omega) = 1 - e^{-j\omega}$$



$$= e^{j\omega/2} e^{-j\omega/2} - e^{-j\omega/2} e^{-j\omega/2}$$

$$= e^{-j\omega/2} (2j) \sin(\omega/2)$$

$$= \underbrace{2 \sin(\omega/2)}_{\text{real valued}} e^{-j(\omega/2 - \pi/2)}$$

$$= A(\omega) e^{j\phi(\omega)}$$

real valued $A(\omega) = 2 \sin \omega/2$

Amplitude response

Phase resp $\phi(\omega) = -\omega/2 + \pi/2$

Group delay $= \tau(\omega) = -\frac{d}{d\omega}\phi(\omega)$

$= 1/2$

constant Group delay.

CT

$\cos(\omega_0 n) \Rightarrow \cos(\omega_0 t)$

(GLP)
Generalized Linear Phase Systems

Phase response $\phi(\omega) = -(\omega\alpha + \beta)$

$$H(\omega) = A(\omega)\, e^{-j(\omega\alpha+\beta)}$$

$A(\omega)$: real valued amplitude response

Group delay $\tau(\omega) = -\frac{d}{d\omega}\phi(\omega) = \alpha$

(Constant group delay)

- Focus on designing real GLP systems

(Real valued Impulse response)

Consider $g(n)$ real and even signal

$$g(-n) = g(n) = g^*(n)$$

$$G(-\omega) = G(\omega) = G^*(-\omega)$$

even

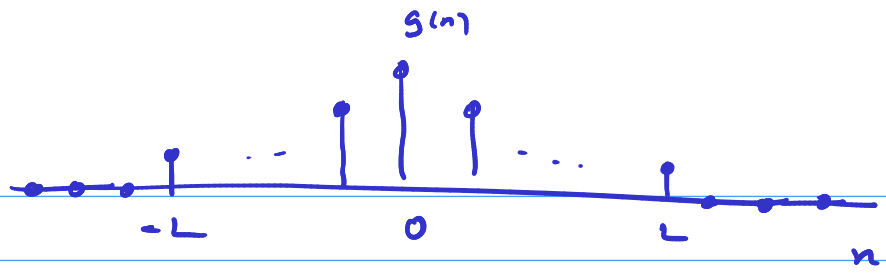
real

$G(\omega)$ is real and even

(Zero phase system)

Suppose $g(n)$ is Finite Impulse Response (FIR)

(duration)



$g(n) = g(-n)$ $\qquad g(n) = 0 \; ; \; |n| > L$

$$G(\omega) = g(0) + g(1)e^{-j\omega} + g(2)e^{-j2\omega} + \cdots + g(L)e^{-jL\omega}$$
$$+ g(-1)e^{j\omega} + g(-2)e^{+j2\omega} + \cdots + g(-L)e^{+jL\omega}$$

$$= g(0) + 2g(1)\cos\omega + 2g(2)\cos(2\omega) + \cdots + 2g(L)\cos(L\omega)$$

(real and even)

Suppose $h(n) = g(n-L)$

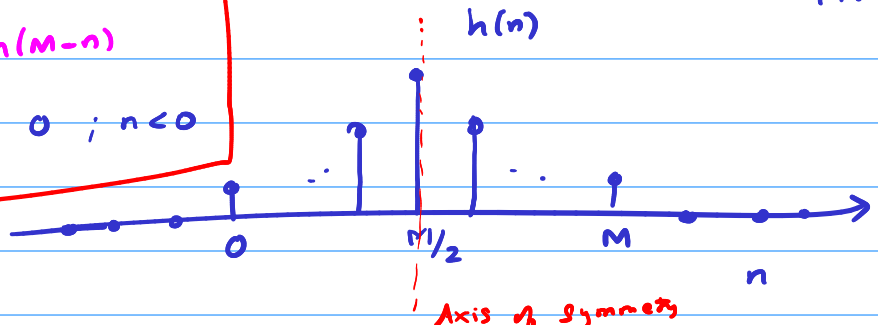
Define $M = 2L$ $\qquad h(n) = g(n - M/2)$

$$H(\omega) = G(\omega)e^{-j\omega M/2}$$

M is even

$h(n) = h(M-n)$

$h(n) = 0 \; ; \; n < 0$



$$H(\omega) = A(\omega)e^{-j\omega M/2}$$

Amp. resp. $\quad A(\omega) = h(M/2) + \sum_{k=1}^{M/2} 2h(M/2 - k)\cos(\omega k)$

Phase resp. $\quad \phi(\omega) = -\omega M/2$

Group delay $\quad \tau(\omega) = M/2$ (Integer)

Type I real GLP FIR filter

Type II GLP FIR Filter

$h(n) = h(M-n)$ (even symmetry)

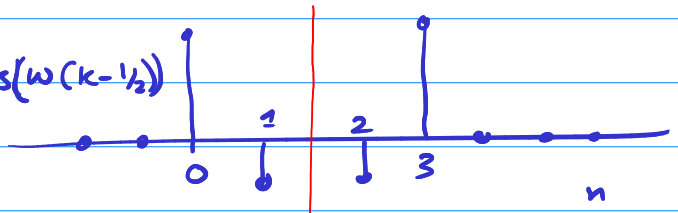
$h(n) = 0$ for $n < 0$ (causal)

M => odd integer

$$H(\omega) = A(\omega) e^{-j\omega M/2}$$

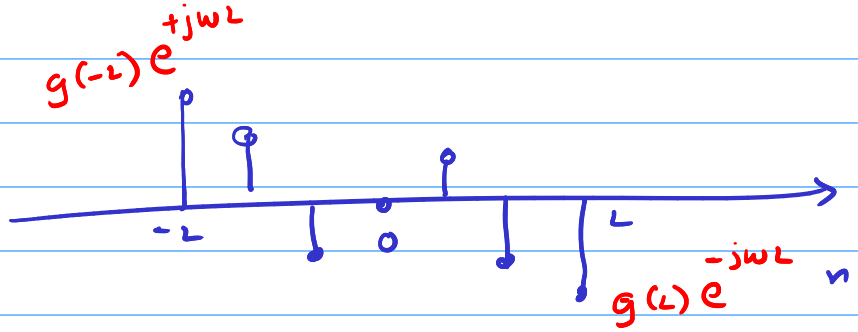
$$A(\omega) = \sum_{k=1}^{(M+1)/2} 2\, h\left(\frac{M+1}{2} - k\right) \cos\left(\omega\left(k - \tfrac{1}{2}\right)\right)$$

Group delay = $M/2$ (non integer)



Recall: If $g(n)$ is real and odd signal

$G(\omega)$ is purely imaginary and odd.



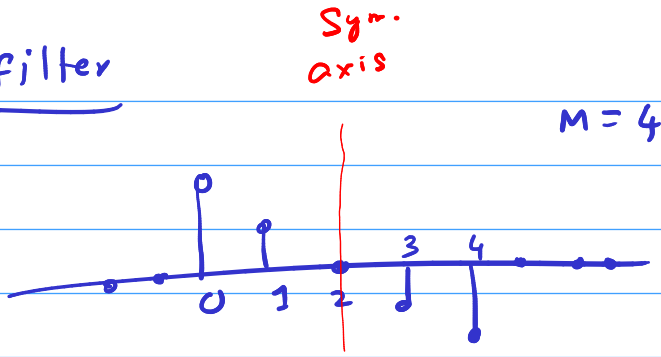
$g(n) = -g(-n) = g^*(n)$

$g(L)\, e^{-j\omega L} + g(-L)\, e^{+j\omega L}$

$- g(L) e^{j\omega L}$

$= -2j \sin(\omega L)$

Type III GLP filter



$h(n) = -h(M-n)$ (odd Sym)

$h(n) = 0$ for $n < 0$

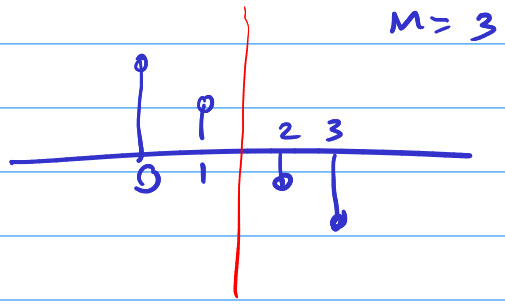
M = even

$$H(\omega) = A(\omega)\, e^{-j(\omega M/2 - \pi/2)}$$

$$A(\omega) = \sum_{k=1}^{M/2} 2\, h(M/2 - k) \sin(\omega k)$$

$\tau(\omega) = M/2$ (integer)

Type IV GLP



$h(n) = -h(M-n)$

$h(n) = 0$ for $n < 0$

M = odd

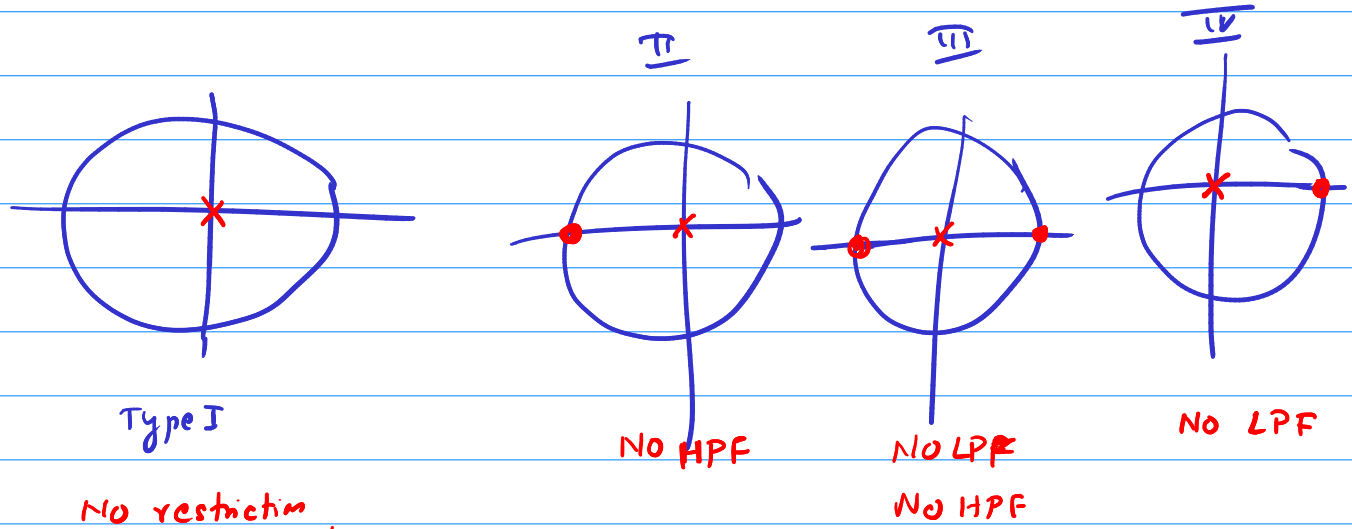
$$A(\omega) = \sum_{k=1}^{(M+1)/2} 2\, h\left(\frac{M+1}{2} - k\right) \sin\left(\omega(k - 1/2)\right)$$

$\tau(\omega) = M/2$ (non integer)

Restrictions on locations of zeros in H(z) due to symmetry.

gain @ $\omega = 0$ $\Rightarrow$ $H(z)\Big|_{@\, z=1} = \sum_n h(n)$

gain @ $\omega = \pi$ $H(z)\Big|_{@\, z=-1} = \sum_n h(n)(-1)^n$



No restriction

Additional Constraints on zeros

$h^*(n) = h(n) = \pm h(M-n)$

$\downarrow \quad \downarrow \quad \downarrow$

$H(z^*) \quad H(z) \quad \pm z^{-M} H(1/z)$

$\Downarrow$

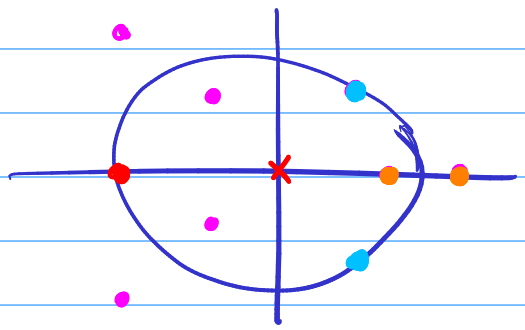
zero at z_0

$0 = H(1/z_0) \Leftarrow H(z_0) = 0 \Rightarrow H^*(z_0^*) = 0$

If z_0 is a zero z_0^* is also a zero

$1/z_0$ is also zero $1/z_0^*$ is also a zero.

An example of pole zero plot for Type II GLP



Final Remark

- Symmetry is not necessary for Linear phase.

Example $H(\omega) = \begin{cases} e^{-j\omega\alpha} & ; |\omega| < \omega_0 \\ 0 & ; \omega_0 \le |\omega| \le \pi \end{cases}$ LPF with linear phase

$H(\omega)$: linear phase

$$\Rightarrow h(n) = \frac{\sin(\omega_0(n-\alpha))}{\pi(n-\alpha)}$$

If α is integer then $h(n)$ is symmetric

Suppose $\omega_0 = 0.5$, $\alpha = 2.93$ } $h(n)$ will not be symmetric

(Sampling Sinc with an offset)